

Senior High School



Learning Activity Sheet for General Mathematics

Quarter 3

Unit
8

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LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	3
Lesson No.:	8.1	Date:	
Lesson Title/Topic:	Estimating and Measuring Lengths and Distances		
Name:		Grade & Section:	

I. Activity No. 1	Practicing the Skills on Estimating and Measuring Lengths and Distances																																																
II. Objective:	At the end of the activity, learners will be able to apply estimation and measurement skills to solve real-life problems involving length, distance, and cost, showing complete and accurate solutions using appropriate units and rounding rules.																																																
III. Materials Needed:	pen calculator																																																
IV. Instructions:	Read and follow the instructions carefully. Work collaboratively with your group in completing each part of the activity. Use appropriate tools and units of measurement when required. Record all your observations and answers neatly in the activity sheet provided. Be sure to compare your estimates with the actual measurements and reflect on your findings through the given processing questions. <i>Part A: Estimation</i> - Each group shall select five (5) objects within the classroom or nearby areas to be estimated. - Using only familiar items inside your bag (such as a ballpen, notebook, etc.) estimate the length or distance of each chosen object. - Record all estimated lengths or distances in the activity sheet provided. <table><tr><th>Object</th><th>Estimated Measurement</th><th>Unit Used (Familiar Object)</th><th>Remarks</th></tr><tr><td>1.</td><td></td><td></td><td></td></tr><tr><td>2.</td><td></td><td></td><td></td></tr><tr><td>3.</td><td></td><td></td><td></td></tr><tr><td>4.</td><td></td><td></td><td></td></tr><tr><td>5.</td><td></td><td></td><td></td></tr></table> <i>Part B: Measurement</i> - Using appropriate measuring instruments such as a ruler, measuring tape, or trundle wheel, measure the actual length or distance of each object selected. - Record the actual measurements beside the corresponding estimates. <table><tr><th>Object</th><th>Estimated Measurement</th><th>Actual Measurement</th><th>Measuring Tool Used</th></tr><tr><td>1.</td><td></td><td></td><td></td></tr><tr><td>2.</td><td></td><td></td><td></td></tr><tr><td>3.</td><td></td><td></td><td></td></tr><tr><td>4.</td><td></td><td></td><td></td></tr><tr><td>5.</td><td></td><td></td><td></td></tr></table>	Object	Estimated Measurement	Unit Used (Familiar Object)	Remarks	1.				2.				3.				4.				5.				Object	Estimated Measurement	Actual Measurement	Measuring Tool Used	1.				2.				3.				4.				5.			
Object	Estimated Measurement	Unit Used (Familiar Object)	Remarks																																														
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	<i>Part C: Comparison and Reflection</i> 1. Compute the difference between the estimated and actual measurements. 2. Reflect on your results by answering the processing questions below.				
	Object	Estimated Measurement	Actual Measurement	Difference	More Accurate Estimation (Yes/No)
	1.				
	2.				
	3.				
	4.				
	5.				
V. Synthesis	Answer the following questions: 1. Which of your estimates were closest to the actual measurements? 2. What familiar object gave you the most accurate estimates? Why? 3. What did you learn about the importance of using standard units of measurement? 4. How can estimation be useful in daily life situations? 5. How did teamwork help in completing this activity?				

I. Activity No. 2	Estimating Lengths Around You
II. Objective:	At the end of the activity, you will be able to apply estimation and measurement skills to solve real-life problems involving length, distance, and cost, showing complete and accurate solutions using appropriate units and rounding rules.
III. Materials Needed:	tape measure or meter stick pen and paper calculator
IV. Instructions:	Work as a group. Perform the required tasks by making estimates first, then measuring accurately. Task 1: Measuring the Whiteboard - You are asked to estimate the length of your Use your arm length or handspan to measure how wide the whiteboard is. - Record how many arm lengths or handspans it takes to cover the whole whiteboard. - After estimating, use a ruler or meter stick to find the actual measurement. Questions: - Was your estimate close to the actual length? - What might have caused the difference? Task 2: Measuring the Distance to the Flagpole - Your group will estimate the distance from your classroom door to the flagpole in the school ground. - Use footsteps to estimate how far the flagpole is from your classroom. - Count your steps as you walk naturally toward the flagpole. - Compare your estimated distance to the actual measurement using a tape measure. Question: - How accurate was your estimation using footsteps?

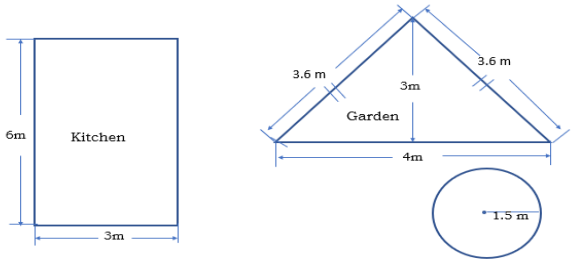
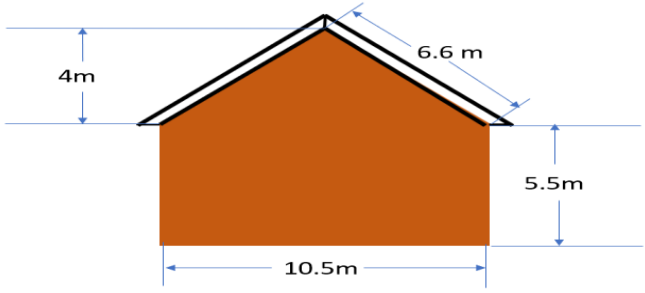
	Task 3: Measuring the classroom door - You are asked to estimate the length of a bench found in the classroom door. - Use a handspan or a notebook as your measuring tool. - Count how many handspans or notebooks long the bench is. - Compare your estimated distance to the actual measurement using a meterstick. Question: - Which tool (handspan or notebook) gave a more accurate result? Why?
V. Synthesis:	Answer the question below. What do we learn about the role of estimation vs. accurate measurement?

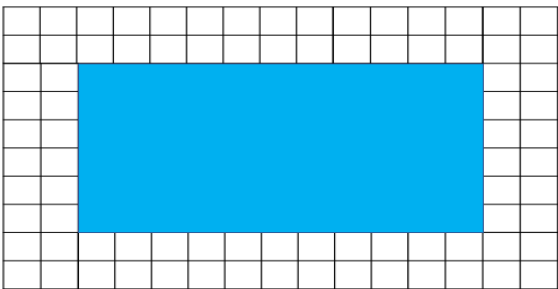
I. Activity No. 3	Assessing Mastery
II. Objective:	At the end of the activity, you will be able to apply estimation and measurement skills to solve real-life problems involving length, distance, and cost, showing complete and accurate solutions using appropriate units and rounding rules.
III. Materials Needed:	pen paper calculator
IV. Instructions:	Read and answer each item. - Which of the following best defines estimation? A. Converting one unit to another B. Measuring an object using a ruler C. Calculating the exact length of an object D. Making an approximate judgment of a value or quantity - If one meter is approximately the distance from your shoulder to your fingertips, what does this show? A. The use of a trundle wheel B. A standard measuring tool C. The length of a small object D. An example of estimation using a familiar reference - Which of the following is the best tool to measure the length of a football field? A. Ruler B. Tape measure C. Trundle wheel D. Odometer - A ribbon costs ₱20 per meter. If you need 4.5 meters, how much will you pay? A. ₱80 B. ₱85 C. ₱90 D. ₱100 - An odometer reading increased from 25,320 km to 25,645 km after a trip. How far did the car travel? A. 225 km B. 315 km C. 325 km D. 335 km - Which of the following situations best shows when estimation is useful? A. Exact measurement is always required B. We want to save time and get a quick idea of size or distance C. We are building machines that require precision D. We cannot use measuring tools at all - A dressmaker needs 2.5 meters of fabric for one dress. If she makes 3 dresses, how many meters of fabric are needed in total?

	A. 5.5 m B. 6.5 m C. 7.5 m D. 8.5 m 8. Why might two learners measuring the same desk using hand spans get different results? A. The desk changed size B. They used different rulers C. Their hand spans are not of equal length D. One student miscounted the desk's corners 9. Which statement best explains the advantage of using standard measuring tools over body parts? A. They are faster to use. B. They are less expensive. C. They are smaller and lighter. D. They are more accurate and consistent. 10. You are asked to measure the length of a school playground but do not have a measuring tape. What is the best strategy to obtain an approximate measurement? A. Guess the length by looking at it B. Wait until someone brings a ruler C. Use your notebook to measure repeatedly D. Use large steps or a trundle wheel if available to estimate the distance.
V. Synthesis:	Answer the following questions: 1. Have you completed all the tasks correctly? 2. Which items did you find easy, and which did you find challenging? 3. What common errors did you make? 4. How will you address your misconceptions?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	3
Lesson No.:	8.2	Date:	
Lesson Title/Topic:	Calculating Area and Perimeter for Practical Purposes		
Name:		Grade & Section:	

I. Activity No. 1	Practice Problems on Area and Perimeter
II. Objectives:	At the end of the activity, you should be able to: 1.. calculate the area and perimeter of rectangular, square, triangular, circular and composite residential spaces. 2. solve contextual problems involving room renovation and garden lay-outing
III. Materials Needed:	pen paper calculator
IV. Instructions:	Analyze and solve the given problems. 1. Maria is improving her house by tiling her kitchen floor, building a flower garden, and installing a circular fountain in the yard. • The kitchen floor is a rectangle measuring 6 meters by 3 meters. • The flower garden is triangular with a base of 4 meters and a height of 3 meters. The other two sides of the triangle are 3.6 meters each. • The fountain has a radius of 1.5 meters.  Answer the following questions: a. What is the area of the kitchen floor that needs to be tiled? b. What is the area of the triangular flower garden? c. What is the perimeter of the garden? d. What is the area and circumference of the fountain? e. Why is it useful for homeowners to know area and perimeter when budgeting for home improvements? 2. Suppose you are painting one side of a house (refer to the illustration below). a. How many square meters of wall will be painted? b. What is the total perimeter of the space to be painted? 

	3. The diagram below shows the design of a tile border around a rectangular swimming pool that measures 3meters by 5.5meters. Each tile is a square measuring 0.5meter in each side.  Questions: - What is the area of the pool? - What is the area of the pool and the tiles? - Using the model above, how many tiles are needed if the pool is 5meters long and 8meters wide
V. Synthesis	Answer the question briefly. Which problem did you find the most challenging, and what strategies or steps did you use to solve it?

I. Activity No. 2	Home Makeover
II. Objectives:	At the end of the activity, you should be able to: - calculate the area and perimeter of rectangular, square, triangular, circular and composite residential spaces. - solve contextual problems involving room renovation and garden lay-outing
III. Materials Needed:	pen paper calculator
IV. Instructions:	Form Your Team Work in groups of 3-4. Each group act as a mini home renovation team. Choose Your Project Select one project scenario (or use the one assigned by your teacher): Room renovation (installing tiles, painting walls, or adding molding). Garden layout (adding a fence, planting grass, or designing flower beds). Draw Your Plan On manila paper or graphing paper, create a simple floor plan or garden layout. Label the sides with measurements (length and width). Be creative but realistic. Do the Calculations - Find the perimeter of your layout. (Use this for things like fences, borders, or wall moldings) - Find the area of your layout. (Use this for things like tiles, paint, or grass) - Show your complete solution clearly. Apply the Price List - Use the price list provided by your teacher (e.g., cost per meter of fence, cost per square meter of tiles). - Compute the total cost of your renovation project. Prepare Your Presentation - Present your group's output to the class. - Be ready to explain: - Your drawing and calculations. - How perimeter and area were used in your project. - Why knowing these measurements are important in real-life renovations and budgeting?

V. Synthesis	Answer the following questions: 1. How perimeter and area affect material planning and costs 2. What are challenges you've faced in measuring, calculating, or budgeting 3. How this activity can be applied in your own homes.

I. Activity No. 3	Assessing learning gain
II. Objectives:	At the end of the activity, you should be able to: 1.. calculate the area and perimeter of rectangular, square, triangular, circular and composite residential spaces. 2. solve contextual problems involving room renovation and garden lay-outing
III. Materials Needed:	pen paper calculator
IV. Instructions:	Answer the following questions. Show your solutions for computation items and use neat sketches for drawing items. 1. A rectangular living room has a length of 8 m and a width of 6 m. If tiles cost ₱300 per square meter, how much will it cost to tile the entire floor? 2. Mr. Cruz is renovating his backyard. He plans to build a rectangular patio that measures 10 meters long and 6 meters wide. At the center of the patio, he will install a circular garden with a radius of 2 meters. a. What is the area of the patio? b. What is the perimeter of the patio? c. What is the area of the circular garden? d. How much space (area) remains in the patio for tiles after placing the circular garden? e. If tiles cost ₱250 per square meter, how much will Mr. Cruz spend on tiling the remaining area of the patio? 3. Maria wants to install curtain rods along the four walls of her rectangular living room. The room measures 6 meters long and 4 meters wide. How many meters of curtain rods does Maria need? 4. A family plans to put grass on their rectangular garden that measures 10 meters long and 8 meters wide. What is the total area of the garden that will be covered with grass? 5. A rectangular kitchen has a length of 5 meters and a width of 3 meters. The family wants to put tiles on the floor, and each square meter of tile costs ₱400. How much will they spend for the tiles?
V. Synthesis	Answer the question briefly. Which problem did you find the most challenging, and what strategies or steps did you use to solve it?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	3
Lesson No.:	8.3	Date:	
Lesson Title/Topic:	Calculating the Volume and Capacity and Its Application on Cost Estimation		
Name:		Grade & Section:	

I. Activity No. 1	Be a Contractor
II. Objective:	At the end of the activity, you should be able to apply the concepts of volume and capacity to real-life situations by computing, converting, and interpreting measurements to solve practical problems involving space and cost.
III. Materials Needed:	pen paper calculator
IV. Instructions:	- Form a group of 4–5 classmates. - Assign your roles: solver, writer, checker, presenter. - Work together to solve the problems that will assign to your group. - The solver solves the problems carefully. - The checker checks another group's answers. - The writer prepares your group's output and summary. - The presenter presents your work to the class. Problem Set 1. Home Improvement Project Filling a Water Drum A cylindrical water drum has a radius of 0.3 m and a height of 1 m. If water costs ₱50 per cubic meter, how much will it cost to fill the drum completely? Painting a Water Tank A cylindrical water tank has a radius of 0.8 m and height of 2 m. You need to paint the outside surface (including the top but not the bottom). Paint costs ₱100 per square meter. Find the total cost of painting the tank. (Use $Surface\ Area = 2\pi r^2 + 2\pi rh$) Constructing a Rectangular Fish Pond A rectangular pond is 4 m long, 3 m wide, and 1.2 m deep. It will be lined with cement 5 cm thick. Cement costs ₱2,000 per cubic meter. Compute: - The volume of cement needed. - The total cost. Problem Set 2. School Facility Project Lower Box Volume A rectangular flower box measures 1.2 m long, 0.5 m wide, and 0.4 m deep. If soil costs ₱300 per cubic meter, find the total cost to fill one box. Rainwater Collector A cylindrical drum with a radius of 0.4 m and height of 1.2 m is used to collect rainwater. If the drum costs ₱250 per cubic meter of material, estimate the total cost. Cylindrical Water Tank with Dome A tank has a cylindrical base (radius = 1.5 m, height = 3 m) topped with a hemispherical dome. Material costs ₱400 per cubic meter. Find the total volume and cost of the tank. Problem Set 3. Business Project Ice Cream Container A cylindrical container has a radius of 7 cm and a height of 10 cm. If 1 liter of ice cream costs ₱250, find the cost to fill 10 containers. ($1,000\ cm^3 = 1\ liter$) Milk Carton Production

	A rectangular milk box measures $8\text{ cm} \times 6\text{ cm} \times 15\text{ cm}$. If each carton costs ₱0.30 per 100 cm^3 to produce, find the cost of producing 20 cartons. 3. Water Delivery Truck A cylindrical water tank (radius = 0.6 m, length = 3 m) delivers water at ₱0.15 per liter. How much will the driver earn per trip if the tank is full? ($1\text{ m}^3 = 1,000\text{ L}$) Problem Set 4. Construction and Design Project 1. Wooden Box Storage A rectangular wooden box measures $0.5\text{ m} \times 0.4\text{ m} \times 0.3\text{ m}$. Wood costs ₱1,200 per cubic meter. Find the cost of wood to make one box. 2. Concrete Pillars A house needs 4 cylindrical pillars, each with a radius of 0.25 m and a height of 3 m. Concrete costs ₱2,500 per cubic meter. Find the total concrete cost for all pillars. 3. Rectangular Pool with Semi-Circular End A swimming pool is 8 m long, 4 m wide, and 1.5 m deep, with a semicircular end (radius = 2 m). Water costs ₱35 per cubic meter. Find the total cost to fill the pool.
V. Synthesis	Answer each question briefly. - What strategies did your group use to solve the problems effectively? - How did working in groups help you understand the concepts better? - What challenges did your group encounter, and how did you overcome them? - How did reviewing another group's work (as a Checker) improve your understanding? - How can you apply what you learned today to real-life situations?

I. Activity No. 2	Mini Measurement Challenge
II. Objective:	At the end of the activity, you should be able to apply the concepts of volume and capacity to real-life situations by computing, converting, and interpreting measurements to solve practical problems involving space and cost.
III. Materials Needed:	- Measuring tape, ruler, or meter stick - Calculator or mobile calculator app - Notebook or data sheet - Local prices of materials (water, paint, tiles, or containers) - Conversion chart
IV. Instructions:	- Select a small container or space (e.g., a water drum, box, bottle, or small tabletop area). - Measure its dimensions—length, width, and height (or radius and height if cylindrical). - Compute the volume using the correct formula: - Convert the volume to an appropriate unit for capacity (e.g., liters). - Estimate the cost of filling or covering it using local material prices (e.g., water, paint, or storage contents). - Record your work clearly in your notebook or data sheet. - Sketch the object or space with labeled dimensions to visualize your calculations.
V. Synthesis	Answer each question. - How does knowing volume and capacity help in planning costs? - What challenges did you encounter in measuring or computing?

	3. How would your estimate change if the material price increases?
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I. Activity No. 3	Assessment on Volume, Capacity, and Cost Estimation
II. Objective:	At the end of the activity, you should be able to apply the concepts of volume and capacity to real-life situations by computing, converting, and interpreting measurements to solve practical problems involving space and cost.
III. Materials Needed:	pen paper
IV. Instructions:	Analyze and solve each problem. 1. Cylindrical Drum Problem: A cylindrical drum has a diameter of 1 m and a height of 1.5 m. - Calculate its capacity in liters - If water costs ₱35 per kiloliter, estimate the total cost to fill the drum. 2. Cube Problem: A cube has an edge of 20 cm. - Find its volume in liters. - If the cube will be filled with rice costing ₱50 per liter, calculate the total cost. 3. Tank Conversion & Reasoning: A water tank has a volume of 2 m³. - How many liters of water can it hold? - Explain why converting from m³ to liters is important when calculating water costs. 4. Box Volume and Unit Conversion: A box has a volume of 45,000 cm³. - Convert this volume to liters. - If a product costs ₱40 per liter, determine the total cost to fill the box. 5. Conceptual Understanding & Analysis: Which statement about cubic meters is correct, and explain your choice: 1 m³ = 100 L 1 m³ = 1,000 L 1 m³ = 10,000 L
V. Synthesis	Response to the question below. Which problems were easy or challenging and how they can apply these skills in everyday life?

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	Quarter:	3
Lesson No.:	Unit 8	Date:	
Lesson Title/Topic:	Performance Task		
Name:		Grade & Section:	

I. Activity No. 1	Project Bahay at Paaralan: Practical Measurement and Cost Planning
II. Objective:	At the end of the activity, you should be able to apply geometric and trigonometric concepts to plan, measure, and estimate materials for a practical project in your local environment, demonstrating accuracy, creativity, and real-life problem-solving skills.
III. Materials Needed:	Measuring tape, ruler, or meter stick Calculator or mobile calculator app Notebook or data sheet Local prices of materials (water, paint, tiles, or containers) Conversion chart
IV. Instructions:	Goal: You will apply geometric and trigonometric concepts to plan, measure, and estimate materials for a practical project in your local environment, demonstrating accuracy, creativity, and real-life problem-solving skills. Role: You are a project planner and designer tasked to improve a space or create a three-dimensional object, considering measurement, design, and cost estimation. Audience: Your teacher, classmates, and community members who will see and evaluate your design and planning output. Situation: You will conduct a mini-measurement project in your school or home, choosing a space or object that can be redesigned, constructed, or organized. You will measure, compute, and estimate materials needed for improvement or construction using local prices. Product/Performance: Select a project from the following options: Project 1: “Ayos Tayo sa Silid!” – Classroom redesign (flooring, painting, or furniture layout) Project 2: “Bahay Improvement sa Barangay” – Home or small community space (kitchen, garden, terrace) Project 3: “Tindahan sa Eskwelahan” – Designing a storage or packaging container Measure the dimensions accurately (length, width, height, radius) using rulers, tape measures, or meter sticks. Compute: - Surface area (for painting, covering, or designing materials) - Volume and/or capacity (for storage, container design, or filling purposes) Estimate costs based on local material prices for paint, tiles, containers, or other materials.

5. **Present** your output in a clear, organized, and visually appealing format (poster, report, or digital presentation) including:
- Labeled sketches or diagrams
 - Measurement tables
 - Computation and cost estimate
 - Reflection connecting math concepts to real-life application

Standards:

By the end performance task, you will **apply geometric and trigonometric concepts to solve real-life measurement problems** such as estimating dimensions, designing, and constructing three-dimensional objects, demonstrating accuracy, creativity, and practical understanding of shapes and volumes.

Scoring Guide:

Criteria	Excellent (4)	Proficient (3)	Developing (2)	Beginning (1)
Accuracy of Measurements & Computations 30%	All measurements are precise; surface area, volume, and capacity calculations fully correct with correct formulas and units.	Minor errors in measurements or computations; formulas mostly correct.	Some formulas missing or applied incorrectly; measurements partially accurate.	Major errors or missing computations; measurements mostly incorrect.
Application of Geometric Concepts 25%	Correctly applies relevant concepts to all aspects of project, including 3D shapes, composite volumes, and real-life measurement problems.	Applies most concepts correctly; minor misconceptions.	Applies some concepts but with errors or incomplete understanding.	Limited or no application of geometric/trigonometric concepts.
Cost Estimation 30%	Provides detailed and accurate cost estimate using local prices; includes proper unit conversions and total cost computation.	Cost estimates mostly correct; minor errors in conversion or totals.	Cost estimate incomplete or contains several errors; conversion missing or incorrect.	Cost estimate missing or largely inaccurate.
Presentation and Creativity 10%	Project is very well-organized, visually clear, and creative; includes labeled diagrams, sketches, and tables.	Organized and clear; some visual elements or labeling incomplete.	Somewhat organized; diagrams/tables incomplete or unclear.	Disorganized; lacks visual support; difficult to understand.

	Reflection and Practical Understanding 5%	Reflection demonstrates deep insight; clearly connects measurements, volume, and cost to real-life application.	Reflection shows basic understanding and some connection to real-life context.	Reflection lacks depth or is only partially connected to the project.	Reflection missing, superficial, or irrelevant.
V. Synthesis	Answer each question. 1. How does knowing geometric and trigonometric concept help in planning costs? 2. What challenges did you encounter in measuring or computing? 3. How would your estimate change if the material price increases?				

